

CS & IT ENGINEERING

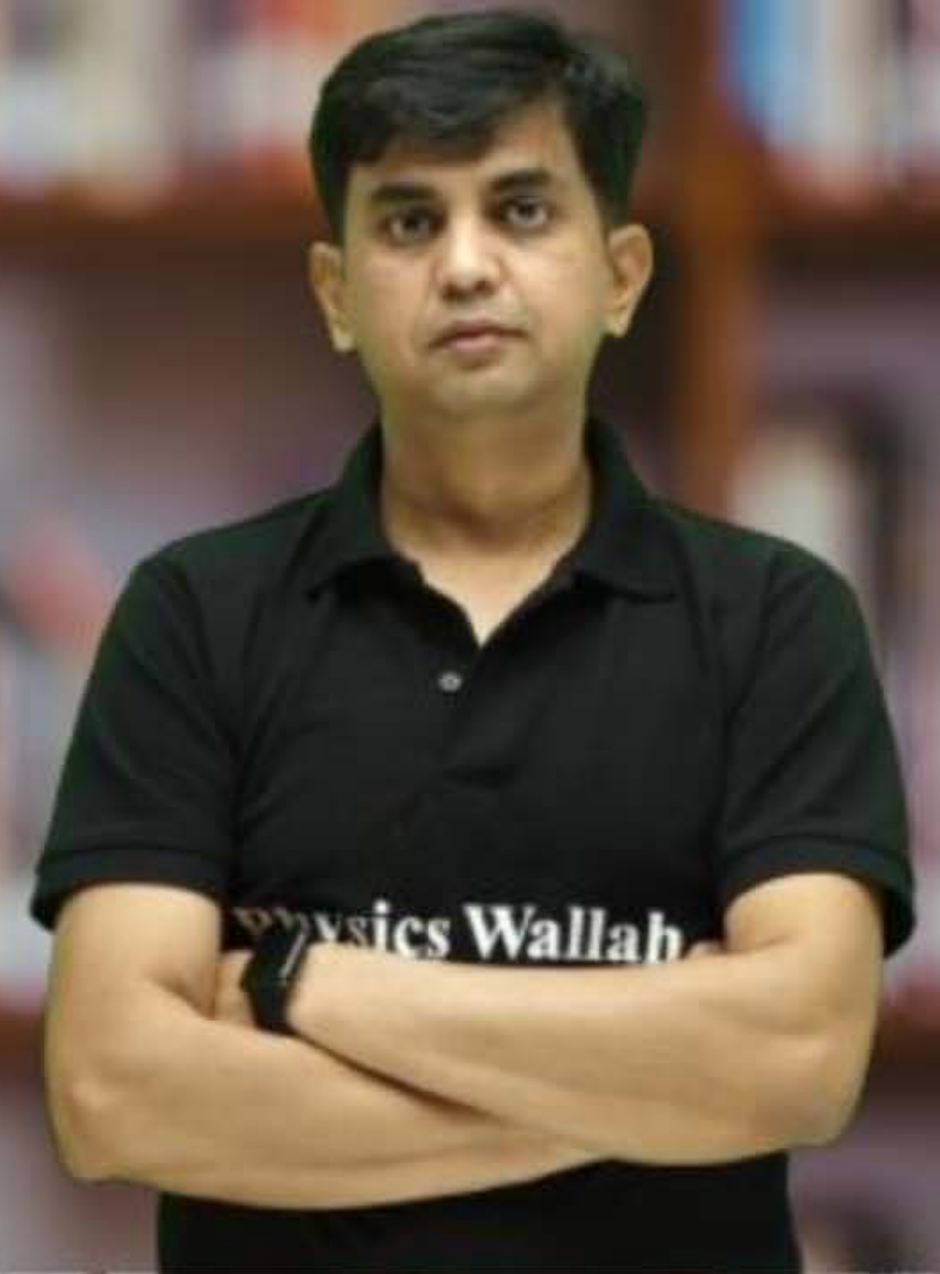


Computer Network 1

IPv4 Address

Lecture No. – 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

Classful IPv4 Address

Topic

Subnetting



Topics to be Covered



Topic

Subnetting



ABOUT ME



Hello, I'm **Abhishek**

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Example 13 :-

Class C network (Network Address : 200 . 200 . 200 . 0), with 3-bit subnetting.

Network Address : 200 . 200 . 200 . 0
Broadcast Address : 200 . 200 . 200 . 255
(Network Directed)
Network Mask : 255 . 255 . 255 . 0

[00000000]
[11111111]

Before Subnetting :-

First Host IP Address : 200 . 200 . 200 . 1
Last Host IP Address : 200 . 200 . 200 . 254
Network Size = $(2^8 - 2) = 254$ hosts

[00000001]
[11111110]

After 3-bit Subnetting :-

Sub-network Address

First Host IP Address

Last Host IP Address

Subnet Broadcast Address
(Sub-network Directed)

24 bit NetID 3 bit SubnetID 5 bit HostID
 : 200 . 200 . 200 . — — — 0 0 0 0 0
 : 200 . 200 . 200 . — — — 0 0 0 0 1
 : 200 . 200 . 200 . — — — 1 1 1 1 0
 : 200 . 200 . 200 . — — — 1 1 1 1 1



First Sub-network Address : 200 . 200 . 200 . 0
First Host IP Address : 200 . 200 . 200 . 1
Last Host IP Address : 200 . 200 . 200 . 30
First Subnet Broadcast Address : 200 . 200 . 200 . 31
(Sub-network Directed)

[00000000]
[00000001]
[00011110]
[00011111]

Second Sub-network Address : 200 . 200 . 200 . 32
First Host IP Address : 200 . 200 . 200 . 33
Last Host IP Address : 200 . 200 . 200 . 62
Second Subnet Broadcast Address : 200 . 200 . 200 . 63
(Sub-network Directed)

[00100000]
[00100001]
[00111110]
[00111111]



Last Sub-network Address : 200.200.200.224
First Host IP Address : 200.200.200.225
Last Host IP Address : 200.200.200.254

Last Subnet Broadcast Address : 200.200.200.255
(Sub-network Directed)

[11100000]
[11100001]
[1111110]
[1111111]

After subnetting: -

Sub-network Mask : 255.255.255.11100000 = 255.255.255.224

Sub-network Size = $[2^5 - 2]$ hosts = 30 hosts per subnet

Network Size = 2^3 subnets per network * 30 hosts per subnet
= 240 hosts per network

After subnetting: -

1st subnet Add. = 200.200.200.0

2nd — // — = — // — . 32

3rd — // — = — // — . 64

4th — // — = — // — . 96

5th 128

6th 160

7th 192

8th subnet Add. = 200.200.200.224



Topic : Sub-network Mask

→ Sub-network Mask (Subnet mask) [32-bits]

NetID field = [All One Bits]

Subnet ID field = [All One Bits]

HostID field = [All Zero Bits]

→ Used to generate subnet address from a given IP address



$$\begin{array}{r}
 200.200.200.50 \\
 255.255.255.0 \leftarrow \text{NM} \\
 \hline
 200.200.200.0 \\
 \text{Network Add.}
 \end{array}$$

$$\begin{array}{r}
 200.200.200.50 \\
 255.255.255.224 \leftarrow \text{SNM} \\
 \hline
 200.200.200.32 \\
 \text{Subnet Add.}
 \end{array}$$



$$\begin{array}{r}
 50 \rightarrow 00110010 \\
 224 \rightarrow 11100000 \\
 \hline
 32 \leftarrow 00100000
 \end{array}$$

[MCQ]

IIT-M, H.W.

[GATE-2019][2 Mark]



#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☐ A M, N and P all belong to the same subnet
- ☐ B Only N and P belong to the same subnet
- ☐ C Only M and N belong to the same subnet
- ☐ D M, N, and P belong to three different subnets



Topic : Size of subnet



→ Sub-network Size : Maximum number of hosts can be in a sub-network

For IP address (32 bits)

NetID field (x bits)

Subnet ID field (y₁ bits)

HostID field (y₂ bits)

$$\text{Sub-network Size} = [2^{y_2} - 2] \text{ hosts per subnet}$$

↑ ↑

[MCQ]

IT →

[GATE-IT-2004][1 Mark]



#Q. A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet?

- ☐ A 14
- ☐ B 30
- ☒ C 62
- ☐ D 126

Ans: C

Subnet Mask :

$$\underline{255.255.255.192} = \underbrace{255.255.255.11000000}_{[NetID+SubnetID]} \underbrace{000000}_{6 \text{ bit HostID}}$$

$$\text{Number of bits for HostID field} = (y_2 \text{ bits}) = 6 \text{ bits}$$

Maximum number of hosts can be in this Sub-network

$$= [2^{y_2} - 2] \text{ hosts per subnet}$$

$$= (2^6 - 2) \text{ hosts}$$

$$= 62 \text{ hosts}$$

[MCQ]

11SC

[GATE-CS-2008][2 Mark]



#Q. If a class B network on the Internet has a subnet mask of 255 . 255 . 248 . 0, what is the maximum number of hosts per subnet?

- ☐ A 1022
- ☐ B 1023
- ☒ C 2046
- ☐ D 2047

Ans: C

Subnet Mask :

$$255.255.248.0 = 255.255.11111000\ 00000000$$

16 bit
NetID
5 bit
subnetting
11 bit host ID

Number of bits for HostID field = (y₂ bits) = 11 bits

Maximum number of hosts can be in this Sub-network

$$= [2^{y_2} - 2] \text{ hosts per subnet}$$

$$= (2^{11} - 2) \text{ hosts}$$

$$= 2046 \text{ hosts}$$



Topic : Number of subnets



For IP address (32-bits)

NetID field (x bits)

Subnet ID field (y_1 bits)

HostID field (y_2 bits)

$$[x + y_1 + y_2 = 32 \text{ bits}]$$

CS-06

Number of subnets = $[2^{y_1}]$ subnets per network

⇒ when subnet ID
all zero and all one
is allowed

$[2^{y_1} - 2]$ subnet per network

⇒ when subnet ID
all zero and
all one is not
allowed

[MCQ]

↓ IIT-B

[GATE-CS-2005][1 Mark]



#Q. An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be :

- ☐ A 255.255.0.0
- ☐ B 255.255.64.0
- ☐ C 255.255.128.0
- ☒ D 255.255.252.0

CS-06
IIT-KGP

Ans: D

Class B Network :-

Number of bits for Network ID field = 16 bits

Number of Subnets = 64 subnets
 [For 64 departments]

Number of bits for Subnet ID field = $\lceil \log_2(\text{Number of Subnets}) \rceil \text{ bits}$

Subnet Mask = 255.255.252.0

$$= \lceil \log_2(64) \rceil \text{ bits} \\ = 6 \text{ bits}$$

= 255.255. 11111100. 00000000
 NetID Subnet ID 6bit HostID 10 bits



Topic : Size of network



→ Network Size : Maximum possible number of hosts can be in a network

For Host IP address (32-bits)

NetID field (x bits)

Subnet ID field (y₁ bits)

HostID field (y₂ bits)

Network Size = Number of Subnets per network * Sub-network size

$$= \left[\underbrace{[2^{y_1}]}_{\text{Number of Subnets per network}} * \underbrace{[2^{y_2} - 2]}_{\text{Sub-network size}} \right] \text{hosts per network}$$



Topic : Reduction in networks size



$$y = y_1 + y_2$$

Network Size (before subnetting)

$$S_1 = [2^{(y_1 + y_2)} - 2] \text{ hosts per network}$$

Network Size (after subnetting) $\Rightarrow$ Subnet ID all zero and all one is allowed

$$S_2 = [2^{y_1}] \text{ subnets per network} * [2^{y_2} - 2] \text{ hosts per subnet}$$

$$= [2^{(y_1 + y_2)} - 2^{(y_1 + 1)}] \text{ hosts per network}$$



Topic : Reduction in networks size



→ Subnet ID field (y_1 bits)

⇒ Subnet ID all zero and all one allowed

Reduction in Network Size (after subnetting)

$$[S_1 - S_2] = [2^{(y_1+1)} - 2] \text{ hosts per network}$$

$$= [2 * 2^{y_1} - 2] \text{ hosts per network}$$

[MCQ]

IIT-KGP, H.W. [GATE-IT-2006] [2 Mark]



#Q. A subnetted Class B network has the following broadcast address : 144 . 16 . 95 . 255. Its subnet mask

- ☐ A is necessarily 255 . 255 . 224 . 0
- ☐ B is necessarily 255 . 255 . 240 . 0
- ☐ C is necessarily 255 . 255 . 248 . 0
- ☐ D could be any one of 255 . 255 . 224 . 0, 255 . 255 . 240 . 0, 255 . 255 . 248 . 0

Example 14 :-

Classless network (Network Address : 75 . 80 . 160 . 0 / 20), with 4-bit subnetting.

Network Address : 75 . 80 . 160 . 0 / 20 101000000 . 000000000

Broadcast Address : 75 . 80 . 175 . 255 / 20 10101111 . 11111111
(Network Directed) 12 bit Host ID

Before Subnetting :-

First Host IP Address : 75 . 80 . 160 . 1 / 20

Last Host IP Address : 75 . 80 . 175 . 254 / 20

Network Size = $(2^{12} - 2)$ hosts = 4094 hosts

After 4-bit Subnetting :-

Sub-network Address : 75.80.1010 ^{20 bit Net ID} ^{4 bit Subnet ID} ^{8 bit Host ID} . 00000000 / 24
 First Host IP Address : 75.80.1010 . 00000001 / 24
 Last Host IP Address : 75.80.1010 . 11111110 / 24
 Subnet Broadcast Address : 75.80.1010 . 11111111 / 24
 (Sub-network Directed)

First Sub-network Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 24

: 75 . 80 . 160 . 0 / 24

First Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 0 0 0 0 0 0 0 1 / 24

: 75 . 80 . 160 . 1 / 24

Last Host IP Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 1 1 1 1 1 1 1 0 / 24

: 75 . 80 . 160 . 254 / 24

First Subnet Broadcast Address : 75 . 80 . 1 0 1 0 0 0 0 0 . 1 1 1 1 1 1 1 1 / 24

(Sub-network Directed) : 75 . 80 . 160 . 255 / 24

Second Sub-network Address : 75.80.1010 0001.00000000 / 24

: 75.80.161.0 / 24

First Host IP Address : 75.80.1010 0001.00000001 / 24

: 75.80.161.1 / 24

Last Host IP Address : 75.80.1010 0001.11111110 / 24

: 75.80.161.254 / 24

Second Subnet Broadcast Address: 75.80.1010 0001.11111111 / 24

(Sub-network Directed) : 75.80.161.255 / 24



8bit Host ID

Last Sub-network Address : 75.80.1010 | | | | .00000000 / 24
: 75.80.175.0 / 24

First Host IP Address : 75.80.1010 | | | | .00000001 / 24
: 75.80.175.1 / 24

Last Host IP Address : 75.80.1010 | | | | .11111110 / 24
: 75.80.175.254 / 24

Last Subnet Broadcast Address : 75.80.1010 | | | | .11111111 / 24
(Sub-network Directed) : 75.80.175.255 / 24

4 bit subnetting



First Sub-network Address : 75.80.160.0 / 24

Second Sub-network Address : 75.80.161.0 / 24

Third Sub-network Address : 75.80.162.0 / 24

Fourth Sub-network Address : 75.80.163.0 / 24

⋮

15th Sub-network Address : 75.80.174.0 / 24

Last (16th) Sub-network Address : 75.80.175.0 / 24

Example 15 :- [H.W.]

Classless network (Network Address : 40 . 50 . 64 . 0 / 19), with 4-bit subnetting.

Network Address : 40 . 50 . 64 . 0 / 19

Broadcast Address : 40 . 50 .
(Network Directed)

Before Subnetting :-

First Host IP Address : 40 . 50 .

Last Host IP Address : 40 . 50 .

Network Size =

Example 16 :- [H.W.]

Classless network (Network Address : 90 . 152 . 0 . 0 / 14), with 5-bit subnetting.

Network Address : 90 . 152 . 0 . 0 / 14

Broadcast Address : 90 .
(Network Directed)

Before Subnetting :-

First Host IP Address : 90 .

Last Host IP Address : 90 .

Network Size =



2 mins Summary



Topic

Subnetting



THANK - YOU

